

Deep learning continues to advance in agriculture, offering accurate plant disease detection to prevent crop losses and support food security. While most studies focus on crops grown in controlled environments, houseplants — commonly kept indoors for aesthetic and health reasons — remain underexplored. Real-world houseplant images often involve complex backgrounds, varied lighting, and irregular angles (Figure 1), and there is a lack of open, well-annotated datasets containing both healthy and diseased samples. These challenges affect model reliability. To address data scarcity, we use classical data augmentation and transfer learning. This study compares two training strategies on a small houseplant image dataset: (1) transfer learning without augmentation, and (2) with classical augmentation. **The goal** is to identify the most effective strategy for houseplant disease classification under limited data conditions.

Figure 1. Houseplant dataset examples



Data & Models

Figure 2. Summary of datasets

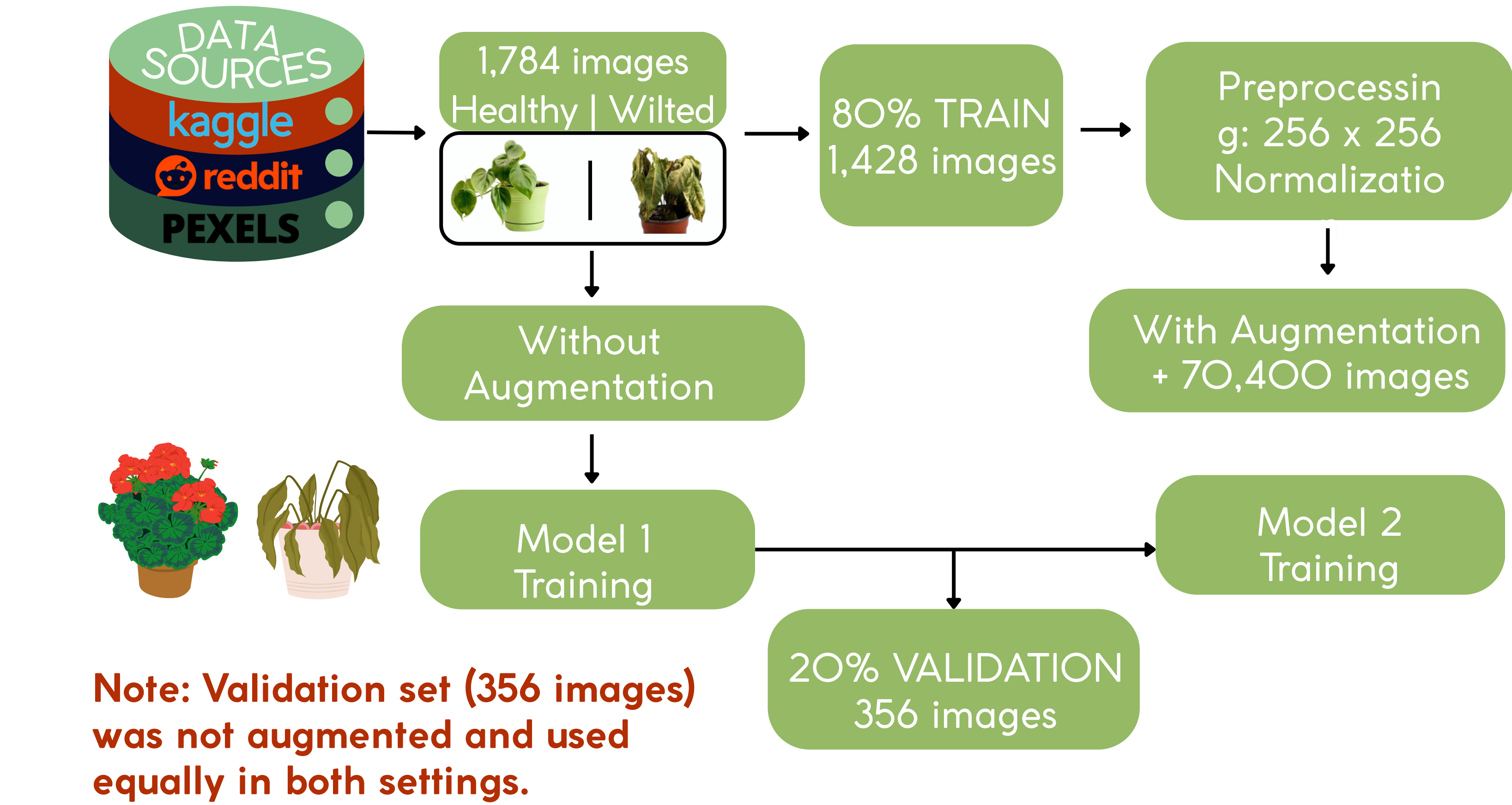
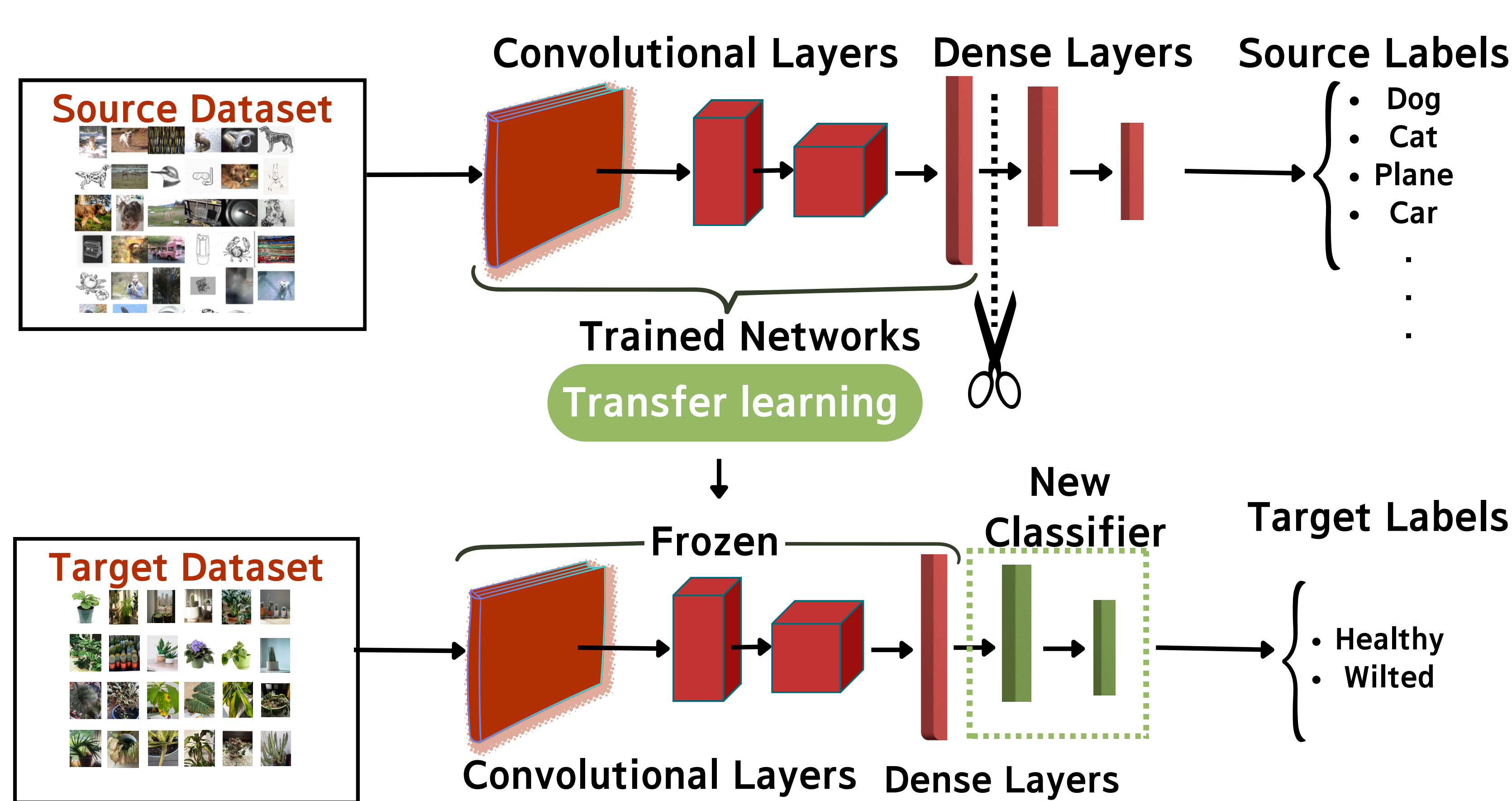


Figure 3. Transfer learning model architecture



Evaluation metrics: accuracy, F1-score, precision, recall. Experimental results reveal distinct impacts of training strategies on model performance (Table 1). MobileNetV2 showed strong generalization with augmentation (accuracy: 0.8399, F1-score: 0.8246), balancing precision and recall. ResNet50 gained the most from augmentation (F1: 0.7560), benefiting from data variation. DenseNet121 performed best without augmentation (accuracy: 0.8989, F1: 0.8966), suggesting its architecture captures sufficient features and may be sensitive to further transformations.

Table 1. Performance comparison of transfer learning models

Transfer learning models	Training Strategy	Accuracy	F1 - score	Precision	Recall
1.MobileNetV2	Original datasets(no augmentation)	0.8118	0.7744	0.9664	0.6461
MobileNetV2	With Augmentation	0.8399	0.8246	0.9116	0.7528
2. ResNet50	Original datasets(no augmentation)	0.6236	0.6278	0.6209	0.6348
ResNet50	With Augmentation	0.7444	0.7560	0.7231	0.7921
3. DenseNet121	Original datasets(no augmentation)	0.8989	0.8966	0.9176	0.8764
DenseNet121	With Augmentation	0.8427	0.8418	0.8466	0.8371

Discussion & Conclusion

This study conducted a comparative analysis of three transfer learning architectures — **MobileNetV2**, **ResNet50**, and **DenseNet121** — under **two data expansion strategies**: with and without classical data augmentation. Results show that **MobileNetV2** and **ResNet50** benefit from classical data augmentation, improving accuracy, F1-score, and recall, indicating that both lightweight and residual architectures gain robustness from diverse input conditions. These variations help mitigate overfitting and enhance generalization, which is critical when working with limited and heterogeneous data such as houseplant images. In contrast, **DenseNet121** achieved the highest accuracy (0.8989) and F1-score (0.8966) without augmentation, suggesting that its densely connected structure already supports effective feature reuse and learning from small datasets. The application of augmentation in this case led to slight performance degradation, possibly due to disturbance in feature propagation caused by redundant transformations. **In conclusion**, classical augmentation remains a reliable strategy for improving model performance on small, non-standardized datasets, particularly when using MobileNetV2 or ResNet50. However, **DenseNet121**, even without augmentation, yields the best results, making it a strong candidate for houseplant disease classification under constrained data conditions.

Future works

Future work may focus on developing structure-aware generative methods (segmentation-based or diffusion models etc.) and expanding the dataset to include a broader range of houseplant species and disease types under diversity background. Improving the visual quality of generated images plays a crucial role, as it directly affects model performance. Exploring domain adaptation and self-supervised learning techniques may further reduce the reliance on large annotated datasets. In parallel, generative methods should accurately replicate key morphological characteristics of houseplants while incorporating diverse backgrounds, camera angles, and lighting conditions to better reflect real-world scenarios. Lastly, deploying models in practical settings and collecting user feedback (e.g., via mobile or edge devices) will be essential for validating their real-life applicability.